Sphinx Defense Full Stack Engineer - What to Learn

Important: eligibility gap (read first)

This JD requires **U.S. citizenship and the ability to obtain/maintain a TS/SCI security clearance**, plus **2-5 years of professional experience**. Both are hard, non-negotiable eligibility requirements for defense-sector roles that no amount of resume tailoring can close — clearance eligibility is a legal/citizenship status, not a skill. This resume was tailored in full per your standard process, but flag this to yourself before investing further time: confirm your eligibility status before an interview, since this is a structural mismatch rather than a skills gap.

Priority: MUST Learn (Class A keywords added to your resume)

- **Kubernetes** — container orchestration on top of Docker (which you already use genuinely via Vibe's E2B sandbox). Learn the core concepts: pods, deployments, services, and how Kubernetes schedules/scales containers that Docker alone doesn't manage. The official "Kubernetes Basics" interactive tutorial (free, browser-based, ~2 hours) covers the essential 20%.
- **Angular** — you know React deeply; Angular is a different framework philosophy (full framework vs. library, TypeScript-first, dependency injection, RxJS observables). Learn Angular's component/module/service structure and how its two-way data binding differs from React's one-way flow. The official Angular "Tour of Heroes" tutorial is the fastest bridge from your React background.
- **Vue** — lighter lift than Angular since Vue's component model is closer to React's (props/state, single-file components). Learn Vue's reactivity system and the Composition API (its hooks-like pattern) to draw a direct parallel to React Hooks you already know.
- **Automated testing (unit, integration, functional)** — you debug production issues via Sentry and do code reviews, but haven't written automated test suites in your listed projects. Learn: Jest/Vitest for unit tests, React Testing Library for component tests, and the difference between unit/integration/functional test scope. This is a real, addressable gap — prioritize it.

Good to Know (Class B — strengthens your fit)

- **Geospatial visualization (2D/3D real-time)** — listed as preferred, tied to space vehicle mission planning. If you want to build a portfolio piece, a simple Three.js or Mapbox GL visualization would be the fastest way to get hands-on exposure; otherwise this is a specialized nice-to-have, not a blocker.
- **Defense/aerospace domain context** — no specific technical prep needed; if you get an interview, be ready to discuss why mission-critical software (higher reliability bar, more rigorous testing/review) appeals to you.

How this maps to your background

Your React/Next.js full-stack experience, Docker usage (Vibe), and relational database work (PostgreSQL in Vibe and Skimly) map directly onto this JD's core technical asks. The gaps are: (1) Angular/Vue as additional frontend frameworks beyond React, (2) Kubernetes as the orchestration layer above Docker, and (3) formal automated testing, which is the most valuable of the three to close since it's a genuine, addressable skill gap rather than a framework-familiarity question. The eligibility requirement (U.S. citizenship + clearance) remains the primary blocker regardless of technical prep.